

LESSON 3.6

MULTIPLYING NUMBERS EXPRESSED IN SCIENTIFIC NOTATION



LESSON INTRODUCTION

MA.8.NSO.1.5 Add, subtract, multiply, and divide numbers expressed in scientific notation with procedural fluency.

LEARNING TARGETS

- I can multiply numbers expressed in scientific notation.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

WORKING TOWARD

MA.912.NSO.1.1 Extend previous understanding of the laws of exponents to include rational number exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

ESSENTIAL QUESTIONS

- How do you multiply numbers expressed in scientific notation?
- What range of numbers are possible for the mantissa?
- How does the power of 10 change as you move the decimal in the mantissa to the left/right?

LESSON NARRATIVE

Students discover how to multiply numbers expressed in scientific notation. Students are engaged in an error analysis activity, guided and independent practice, and a collaborative game. Through this repetitive practice students develop the knowledge and skills to multiply numbers expressed in scientific notation.

COMPONENT SUMMARY

3.6.1 WARM UP (~5 MIN.)

Review adding and subtracting numbers written in scientific notation

3.6.2 GUIDED INSTRUCTION (10 MIN.)

Discover how to multiply numbers written in scientific notation and error analysis

3.6.3 GUIDED PRACTICE (10 MIN.)

Practice multiplying numbers written in scientific notation

3.6.4 ACTIVITY (15-20 MIN.)

Collaborative game multiplying numbers written in scientific notation

3.6.5 WRAP-UP (~5 MIN.)

Reflect on growth mindset actions used

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- Problem Cards
- Answer Sheet
- Game Boards
- Dice (Number Cubes)

ADDITIONAL PREPARATION

- Prepare cards, answer sheets, and boards for the game in 3.6.4 Activity.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not understand why they should apply the product of powers law and add the powers of ten.
- Students may not understand how to rewrite their answers in scientific notation.

THINGS TO CONSIDER

- Consider reviewing the power of products law.
- Consider displaying a scientific notation anchor chart that provides visuals and written instructions for how moving the decimal in the mantissa to the left/right impacts the power of ten.
- If time is an issue, consider using part of 3.6.4 Activity to help students gain fluency before the unit assessment.

LITERACY AND LANGUAGE ROUTINES

Clarify Critique Connect

In 3.6.2 Guided Instruction students see a sample of student work that shows a common error. Initially, clarify how to multiply numbers expressed in scientific notation. Then, prompt students to critique the sample student's work. Finally, engage students in a discussion on what the mistake is, why the student may have made it, how to correct it, and how to avoid making the mistake in the future.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Students are provided with multiple opportunities to practice multiplying numbers written in scientific notation in 3.6.2 Guided Instruction, 3.6.3 Guided Practice, and 3.6.4 Activity. This repeated practice help students develop the knowledge and skills to accurately and efficiently solve the problem.

DIFFERENTIATE INSTRUCTION

The 3.6.4 Activity provides an auditory and kinesthetic entry points as students discuss how to multiply numbers written in scientific notation with their partners and physically manipulate the cards and game board.

Problem #: 31

Answer:

(3.024×10^{19})

Work:

$(9.45 \times 3.2)(10^9 \times 10^9) = 30.24 \times 10^{18}$

3.6.1 WARM-UP: BELL WORK

TEACHER MOVES

Use student understanding to inform your instruction and to determine where to add clarity during this and further lessons. Make a list of misconceptions and determine which should be addressed with the whole group and which should be addressed in a small group with those who are struggling to be proficient.



3.6.1 Warm-Up: Bell Work

Add.

1. $(4.79 \times 10^7) + (1.725 \times 10^5)$

$$\begin{aligned} &47.9 \times 10^6 + 0.1725 \times 10^6 \\ &= 48.0725 \times 10^6 \\ &= 4.80725 \times 10^7 \end{aligned}$$

2. $(2.9 \times 10^4) + (2.5 \times 10^4)$

$$5.4 \times 10^4$$

3. $(1.7 \times 10^{-2}) + (1.3 \times 10^{-2})$

$$3.0 \times 10^{-2}$$

4. $(9.9 \times 10^{-7}) + (7.48 \times 10^{-9})$

$$9.9 \times 10^{-7} + 0.0748 \times 10^{-7} = 9.9748 \times 10^{-7}$$



3.6.2 Guided Instruction: Multiplying Numbers Expressed in Scientific Notation

Find the product of each.

1. $(7.8 \times 10^{14})(2.5 \times 10^{16})$

$19.5 \times 10^{30} = 1.95 \times 10^{31}$

2. $(1.2 \times 10^{-9})(3.2 \times 10^{-7})$

3.84×10^{-16}

3. Celia's work for multiplying $(2 \times 10^5)(4 \times 10^5)$ is shown. She did the problem first using scientific notation and then checked her answer using standard notation. She noticed that the answers do not match.

Scientific Notation	Standard Notation
$(2 \times 10^5)(4 \times 10^5)$	$200,000 \times 400,000$
$(2 \times 4)(10^5 \times 10^5)$	$8 \times 10,000,000,000$
8×10^5	$80,000,000,000$

Find Celia's mistake then write her a note to help her learn from her mistake.

Means

I

Start

To

Acquire

Knowledge

Experience

Skills

When working with scientific notation, don't forget to add the exponents when multiplying two or more terms!

3.6.2 GUIDED INSTRUCTION: MULTIPLYING NUMBERS EXPRESSED IN SCIENTIFIC NOTATION

TEACHER MOVES

Students may struggle with why multiplication of numbers expressed in scientific notation does not require that the powers of 10 match. Use the laws of exponents to help students recognize there is no law for adding numeric exponential expressions but there is the Product of Powers Law. This can also be explained multiplying values such as 1000 and 200 where the place value of the 1 and the 2 does not affect the place value of the answer whereas adding 1000 and 200 the place value of the 1 and the 2 is shown in the result.

Consider breaking question 1 into four parts and prompting students to follow the same procedure with subsequent problems:

1. What is 7.8×2.5 ?
2. What is $10^{14} \times 10^{16}$?
3. What is $(7.8 \times 10^{14})(2.5 \times 10^{16})$?
4. How do you express your answer to part 3 in scientific notation?

DIFFERENTIATE INSTRUCTION

For question 3, consider providing students with sentence frames to help them compose their notes. For example:

- Celia incorrectly _____.
- Celia should have _____.

3.6.3 GUIDED PRACTICE: MULTIPLYING NUMBERS EXPRESSED IN SCIENTIFIC NOTATION

DIFFERENTIATE INSTRUCTION

Consider modeling how students can organize their work in a table. For example:

!!!

Scientific Notation $3.1 \times 10^{-9} \times 3.2 \times 10^{-7}$ = Standard Form $\times =$



3.6.3 Guided Practice: Multiplying Numbers Expressed in Scientific Notation

- Determine the product of $(3.1 \times 10^{-9})(3.2 \times 10^{-7})$.
 9.92×10^{-16}
- Check your work by finding the product of the numbers when written in standard form.
 $0.000000031 \times 0.00000032 = 0.00000000000000992$
- Complete the work to find the product, being sure to write your answer in scientific notation.

$$(8.8 \times 10^4)(4.92 \times 10^5)$$

$$= (8.8 \cdot 4.92)(10^4 \cdot 10^5)$$

$$= 43.296 \times 10^9$$

$$4.3296 \times 10^{10}$$



3.6.4 Activity: Give Me Five!

You and your partner will be given a set of problem cards, and a gameboard. Your goal is to get five squares in a row, horizontally, vertically, or diagonally, marked on the gameboard. To mark a square, you need to correctly find the product of a given problem.

Record each problem that you do in the recording sheet. All of the spaces may not be needed before a winner is determined.

Give Me Five! - Recording Sheet

Answers may vary – all answers provided on the Activity Sheet.

Problem #: 11 Answer: 7.872×10^{27} Work: $(9.6 \times 8.2)(10^{13} \times 10^{13})$ $= 78.72 \times 10^{26}$	Problem #: 21 Answer: (1.4812×10^{12}) Work: $(6.44 \times 2.3)(10^6 \times 10^5)$ $= 14.812 \times 10^{11}$	Problem #: 31 Answer: (3.024×10^{19}) Work: $(9.45 \times 3.2)(10^9 \times 10^9) = 30.24 \times 10^{18}$
Problem #: 12 Answer: (2.944×10^{-21}) Work: $(9.2 \times 3.2)(10^{-12} \times 10^{-10})$ $= 29.44 \times 10^{-12}$	Problem #: 22 Answer: (3.96×10^{-36}) Work: $(1.1 \times 3.6)(10^{-18} \times 10^{-18})$ $= 3.96 \times 10^{-36}$	Problem #: 32 Answer: (4.693×10^{-15}) Work: $(2.47 \times 1.9)(10^{-8} \times 10^{-7})$ $= 4.693 \times 10^{-15}$

3.6.4 ACTIVITY: GIVE ME FIVE!

DIFFERENTIATE INSTRUCTION

For students who need differentiation, consider providing them with modified game cards where the mantissa is a whole number or a number to the tenths place.

TEACHER MOVES

While students are working, consider collecting observational data using a rubric. Use the observations to inform what differentiation or small-group instruction can look like. Specifically focus on what students show understanding on the following:

- Multiplying the two mantissas
- Adding the exponents
- Rewriting the answer in scientific notation, $a \times 10^n$ where $1 \leq |a| < 10$.

ENRICHMENT

What steps did you take to multiply the numbers expressed in scientific notation? Do you agree or disagree with your partner's work? How can you check your work?

3.6.4 ACTIVITY: GIVE ME FIVE! (CONTINUED)

DIFFERENTIATE INSTRUCTION

The game provides auditory and kinesthetic entry points as students discuss how to multiply numbers written in scientific notation with their partners and physically manipulate the cards and game board.

Problem #: 13 Answer: (4.488×10^{18}) Work: $(1.32 \times 3.4)(10^{10} \times 10^8)$ $= 4.488 \times 10^{18}$	Problem #: 23 Answer: (1.0504×10^{-9}) Work: $(4.04 \times 2.6)(10^{-6} \times 10^{-4})$ $= 10.504 \times 10^{-10}$ $= 27.225 \times 10^0$	Problem #: 33 Answer: (2.7225×10^1) Work: $(8.25 \times 3.3)(10^{-1} \times 10^1)$ $= 27.225 \times 10^0$
Problem #: 14 Answer: (1.863×10^{16}) Work: $(2.07 \times 9)(10^8 \times 10^7)$ $= 18.63 \times 10^{15}$	Problem #: 24 Answer: (2.184×10^{17}) Work: $(5.46 \times 4)(10^8 \times 10^8)$ $= 21.84 \times 10^{16}$	Problem #: 34 Answer: (1.6704×10^{13}) Work: $(6.96 \times 2.4)(10^7 \times 10^5)$ $= 16.704 \times 10^{12}$
Problem #: 15 Answer: (3.19×10^{-20}) Work: $(1.1 \times 2.9)(10^{-10} \times 10^{-10})$ $= 3.19 \times 10^{-20}$	Problem #: 25 Answer: (5.9319×10^{42}) Work: $(1.521 \times 3.9)(10^{22} \times 10^{20})$ $= 5.9319 \times 10^{42}$	Problem #: 35 Answer: (9.125×10^{20}) Work: $(7.3 \times 1.25)(10^{10} \times 10^{10})$ $= 9.125 \times 10^{20}$
Problem #: 41 Answer: (2.014×10^{27}) Work: $(5.3 \times 3.8)(10^{14} \times 10^{12})$ $= 20.14 \times 10^{26}$	Problem #: 42 Answer: (5.5696×10^{10}) Work: $(4.72 \times 1.18)(10^5 \times 10^5)$ $= 5.5696 \times 10^{10}$	Problem #: 43 Answer: (2.592×10^{-18}) Work: $(2.16 \times 1.2)(10^{-9} \times 10^{-9})$ $= 2.592 \times 10^{-18}$



3.6.5 Wrap-Up: Growth Mindset

Which mindset did I demonstrate?	Choose the best description. <i>Answers will vary.</i>
Did I use the partner discussions as learning opportunities?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I work on problems that challenged me?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I use strategies to 'un-stick' myself when I found the problems difficult?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I put as much effort as I possibly could into the problems?	☐ Never ☐ Sometimes ☐ Usually ☐ Always

3.6.5 WRAP-UP: GROWTH MINDSET

TEACHER MOVES

Consider engaging students in a discussion about the benefits of the growth mindset thinking or action explained in each prompt.